WORLDPROOF Hackathon Judging Guideline

Build it. Break it. Fix it.

1. Purpose of WORLDPROOF

Southern University of Science and Technology (SUSTech) is located in Shenzhen, an innovation-driven city in the Guangdong-Hong Kong-Macao Greater Bay Area. Its motto, "明德求是日新自强," centers on virtue, truth-seeking, renewal, and self-improvement. The SUSTech-BIO team likewise focuses on how synthetic biology can move beyond the lab and respond to real-world needs and constraints. With this aim, the team created and organizes the WORLDPROOF Hackathon at SUSTech. Unlike a conventional project pitch, WORLDPROOF looks beyond novelty and presentation, asking whether a project can function outside the lab and withstand the realities of use.

Before the event, teams prepare approximately 70% of a working Demo. On the morning of the event, teams take part in a one-hour open booth session to showcase their work, exchange feedback, and act as Red Teams by identifying weaknesses in other projects and issuing challenges. Judges do not attend this session. After receiving a New Constraint Card and suggestions from other teams, participants spend the remaining build period fixing their own Demos. Formal judging takes place in the afternoon, when each team gives a short slide presentation to a panel of both on-site and online judges. Teams present the Demo they revised through this real-world stress test and explain how it responds to the issues raised.

The central judging question is:

Can the team identify where a project may fail, learn from external criticism, and redesign it for responsible real-world use?

WORLDPROOF extends the Design-Build-Test-Learn cycle beyond laboratory optimization by introducing stakeholders, infrastructure, accessibility, governance, uncertainty, and unexpected events.

2. Competition Tracks

Teams enter one of two tracks. Both use the same judging criteria.

PROJECT TRACK

Teams bring an existing iGEM or synthetic biology project. They are not judged on biological novelty, experimental performance, patent potential, or business-plan quality. They are judged on how well they define a real-world challenge, respond to external feedback, and prepare the final application for real-world use.

SCENARIO TRACK

Teams work from a fictional scenario supplied by the organizers. The scenarios are inspired by real news and public challenges. Teams build an application, decision tool, monitoring system, communication interface, simulation, or service prototype around the scenario.

Do not treat one track as more advanced than the other. Scores should reflect the quality of reasoning and iteration, not how mature the starting project was.

3. What Judges Review

Judges review only two items:

Teams present both the Change Log and the Final Demo as part of their afternoon slide presentation; judges are not expected to review separate files in advance.

1. Change Log - the team's record of its prioritized challenges and responses. It must include:
- the New Constraint Card issued by the organizers;
- suggestions received from other teams;
- the one or more challenges the team chose to prioritize;
- why those challenges mattered;
- the design decisions made in response;
- any important issue that remains unresolved.
2. Final Demo - the final version presented during judging.

Judges do not separately review the Baseline Demo or Challenge Cards. These may be retained by the organizers to document the event process.

Material	Used For
Change Log	Challenge / 30
Change Log + Final Demo	Fix / 45
Final Demo + team explanation	World Readiness / 25

4. Suggested Team Presentation Format

The one-hour morning booth session is for participant exchange and Red Team challenges only; it is not part of judging. Formal judging takes place in the afternoon, with both on-site and online judges. Each team should receive approximately six minutes: four minutes for its slide presentation and two minutes for judge questions.

Time	Team Content
30 sec	Team name, track, project/scenario, and intended user
60 sec	Challenge synthesis: which constraints and peer feedback did the team prioritize, and why?
120 sec	Final Demo and the main design decisions made during FIX
30 sec	Remaining uncertainty, limitation, or pause/stop condition
120 sec	Judge questions

Teams should not use the judging period for a full iGEM project presentation. Judges may redirect a presentation that spends excessive time on scientific background.

5. Scoring Criteria

Challenge - 30 points

Assess the Change Log. Determine whether the team understood, synthesized, and prioritized one or more meaningful real-world challenges from the New Constraint Card and peer suggestions.

For each prioritized challenge, a strong framing identifies an affected stakeholder, a specific failure mechanism, a credible consequence, and an observable warning signal.

Score	Standard
0-10	Feedback is copied or listed without a clear priority or failure mechanism.
11-20	One or more plausible challenges are selected, but the explanation of stakeholders, consequences, or warning signals is incomplete.
21-30	The team clearly synthesizes and prioritizes one or more specific, credible challenges, explaining the relevant stakeholder, consequence, and warning signal for each.

Fix - 45 points

Assess the Change Log and Final Demo.

A strong Fix makes a meaningful change to the workflow, product logic, monitoring, permissions, communication, accessibility, escalation, or pause/stop mechanism. Visual polish or additional screens alone do not constitute a strong Fix.

Score	Standard
0-15	No meaningful response, or only a cosmetic change with little connection to the prioritized challenges.
16-30	A relevant change is visible, but the reasoning, implementation, testing, or trade-offs are incomplete.
31-45	A substantial, coherent, and testable redesign directly addresses the prioritized challenges and explains key trade-offs.

World Readiness - 25 points

Assess the Final Demo and the team's explanation of its intended context.

Consider whether the final design supports a clear workflow, addresses users and affected non-users, recognizes practical constraints, states important uncertainty, and includes appropriate monitoring and pause/stop conditions.

Score	Standard
0-8	No workable real-world pathway or responsibility boundary; major risks are ignored.
9-17	A workable basic approach, but important users, constraints, monitoring, or uncertainty remain underdeveloped.
18-25	A credible, inclusive, and operationally realistic design with explicit limits, monitoring, escalation, and pause/stop conditions.

6. Using the Score Sheet

1. Locate the team's row using its team number and name.
2. Confirm whether it is in PROJECT TRACK or SCENARIO TRACK.
3. Enter one score for Challenge, Fix, and World Readiness.
4. Add the three scores for a total out of 100.
5. Use the second row to write one concise, actionable comment.

Comments should identify either the strongest design decision or the single most important improvement needed next. Avoid general comments such as "good presentation" or "interesting project."

7. Judging Principles

- Do not judge the Baseline Demo.
- Do not reward biological novelty or project maturity by itself.
- Do not reward presentation polish unless it improves real-world communication or use.
- Score what is demonstrated and explained, not what a team promises to build later.
- Apply the same standards to both tracks.
- Do not reward a challenge for being dramatic; reward it for being credible and useful.
- Do not reward technically detailed suggestions that could enable biological harm.
- Declare any conflict of interest to the organizers before scoring a team.

8. Tie-Breaking

If two teams receive the same total score, compare them in this order:

1. Higher Fix score.
2. Higher Challenge score.
3. Higher World Readiness score.
4. If still tied, judges discuss which team demonstrated the clearest evidence-to-design reasoning.

9. Final Reminder

WORLDPROOF uses adversarial testing as its method and collaboration as its goal. We encourage teams to expose critical assumptions, respond constructively to criticism, and use what they learn to improve designs for the real world.

Thank you for your time and valuable feedback. Your professional judgment and constructive comments help ensure the quality of WORLDPROOF judging and enable teams to turn their on-site Fix into a stronger next iteration.